

Alex Byun

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Final-year Science and Computer Science student building practical AI automation: orchestrating Claude, APIs/MCP connectors, and scripting to turn manual processes into working tools via workflows.

EDUCATION

University of New South Wales - Bachelor of Science and Computer Science

Final year

Relevant coursework: Operating Systems, Computer Networks & Applications, Software Construction, Data Structures & Algorithms, Object-Oriented Design.

TECHNICAL SKILLS

Languages: Python, Bash / shell, HTML, CSS, C, JavaScript / Node.js, Java

AI & Automation: Claude Code (custom skills, hooks, headless orchestration), workflow creation, multi-agent & subagent designs, parallel process management, prompt engineering & versioning, eval harnesses, FFmpeg media pipelines

Agents & skills built: scheduled autonomous cloud agents, a 5-agent parallel roster with group-debate orchestration (AlexOS), 10+ custom Claude Code skills in daily use (session memory, idea ingest, portfolio sync)

MCP & integrations: Zapier MCP (Gmail + Google Calendar automation), Composio MCP (1000+ apps), Telegram Bot API webhooks, Cloudflare Workers + KV, local Stable Diffusion image generation (Fooocus), REST / JSON API wiring & auth

Tools & Platforms: Vercel, Railway (worker + Postgres + Redis), Netlify, Git & GitHub, VS Code, Linux/Unix, Microsoft Office, Excel

Strengths: documentation & process mapping, clear written communication, fast self-directed learning, detail-oriented

PROJECTS

AlexOS - Local Multi-Agent Operating System

github.com/byun-alex/alexos

Node.js, React, WebSocket · multi-agent orchestration, event-driven, capability-sandboxed writes

- Runs five AI agents as parallel processes against my real files, with live status in the browser and cost tracked per run.
- A debate mode has the agents argue a problem over several rounds before a chair writes the agreed answer to shared memory. Agents can only request actions and the server alone writes to disk, so a bad prompt cannot corrupt data.

Content Factory - Multi-Channel AI Video Production Engine

github.com/byun-alex/content-factory

Python, FFmpeg, Stable Diffusion, Composio MCP · staged pipeline, human approval gates, tiered fallbacks

- Turns a style brief and a runtime into a finished multi-hour video by generating the music and the art, then assembling it with FFmpeg. One engine drives several channels through swappable style guides.
- Generation falls back across local GPU, cheap API, or manual, so a missing service pauses a run instead of killing it. Replacing a paid video API with a local overlay cut that step's cost to zero.

Morning Brief Agent - Autonomous, Spec-Driven Daily Agent

github.com/byun-alex/morning-brief-agent

Node.js, cron, MCP (Gmail, Calendar) · spec-as-config, fails loudly on missing sources

- A scheduled cloud agent that reads my calendar, inbox, deadlines and weather each morning, writes a brief, and emails it before the day starts.
- Its whole behaviour is a Markdown spec read at run time, so changing it is an edit rather than a redeploy. If a source is empty or a tool fails, the brief says so plainly instead of inventing.

Instagram Comment-to-DM Automation - Self-Hosted, Live in Production

[my fork](#)

Next.js on Vercel, Postgres, Redis, queue worker on Railway, Meta webhooks, OAuth · split web/worker, atomic claim on insert

- Self-hosted an open-source app ([diwenne/openreply](#), not my application code) to replace a \$14/month SaaS with an uncapped deployment costing about \$5/month. The worker is split out because a queue consumer must stay awake and serverless sleeps.
- Root-caused a live incident on a post at 57,000 views: the duplicate messages came from a second tool still connected to the same account, not from my code. Fixed a retry storm and a webhook/poll race on the way.

More on GitHub: **log-triage** (support triage CLI, 17 tests) · **murmur** (on-device dictation) · **market-data-scanner** (async market-data engine, 49 tests) · **overnight-agent** (verify-gated job runner) · **telegram-idea-capture** (Cloudflare Worker + KV inbox) · **brain-dashboard** (Node 24 PTY root-cause).

EXPERIENCE

AI Trainer - Outlier (Freelance / Contract)

May 2024 – Mar 2026

- Review and evaluate AI-generated responses (text, code, images, design) for accuracy, clarity, and relevance; document errors with consistent, detail-focused written feedback to improve model output.

Customer Service, Social Media & Admin - Woolworths · Oliver Brown · a Shopify business

2021 – 2023

- Handled frontline customer support, bookings, and accurate record-keeping in fast-paced environments, staying calm and methodical under pressure; managed TikTok, Instagram, and YouTube for a small business, using Shopify analytics to adjust ad planning.

ADDITIONAL

Languages: English (native), Korean (proficient), Japanese (intermediate)

Eligibility: Australian citizen · Full driver's licence

Availability: Available now for full-time work; studies part-time and schedules classes outside working hours.